

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

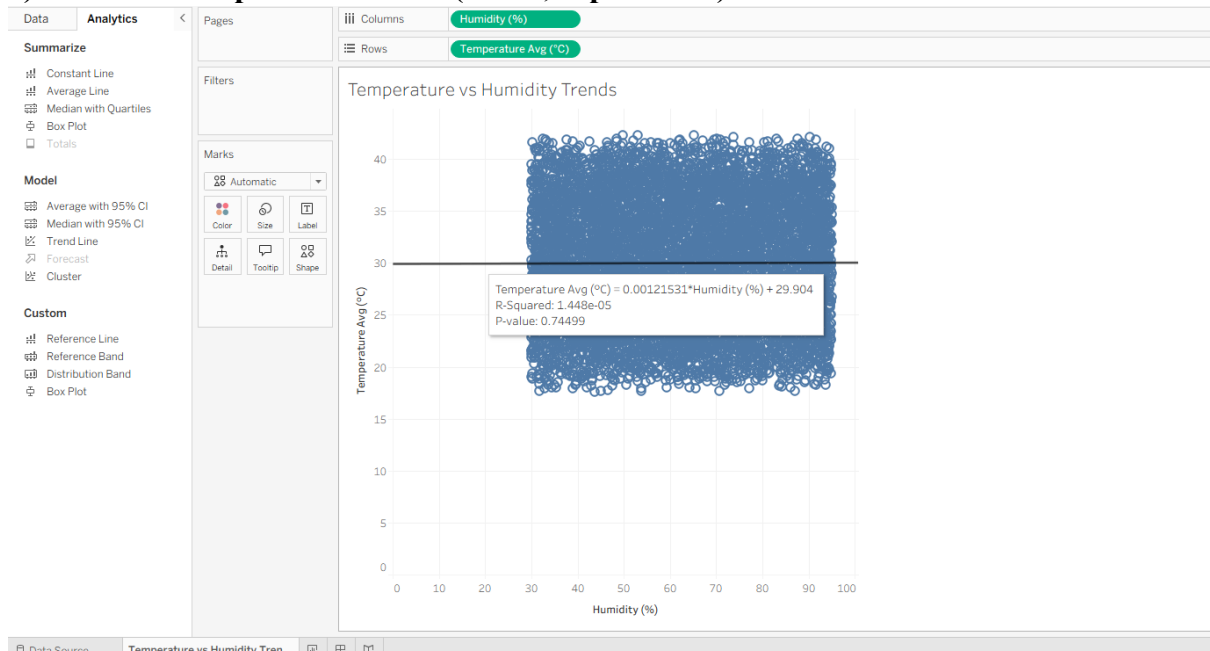
Practical – VIII

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Class:TYIT	Batch:1
Date of Assignment:27-08-2026	Date/Time of Submission: 27-08-2026

Part B – Program

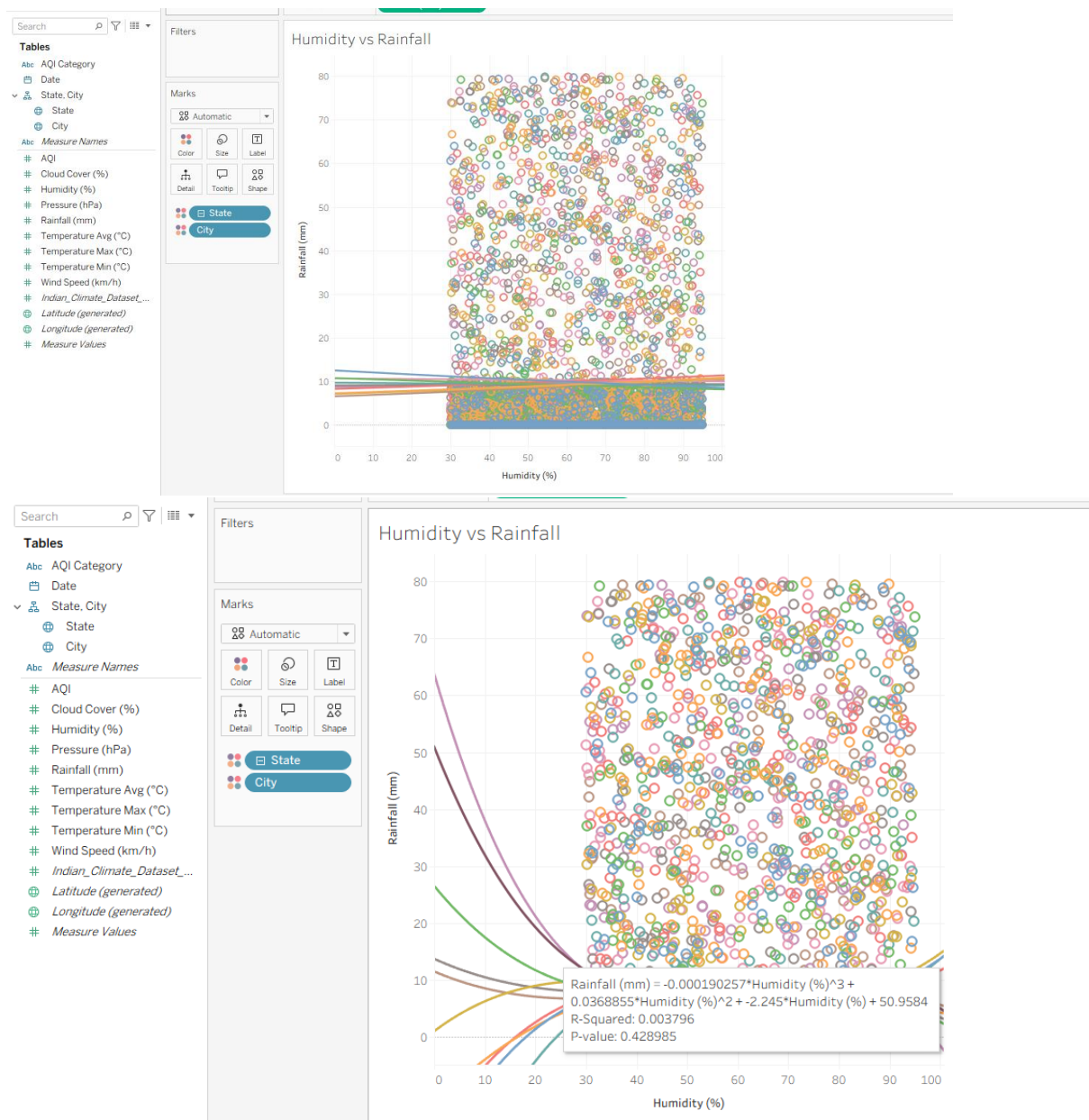
Performing Advanced Analytics using Trends, Clustering, and Forecasting

a) Add and interpret trend lines (linear, exponential)



b) Customize and analyze trend models

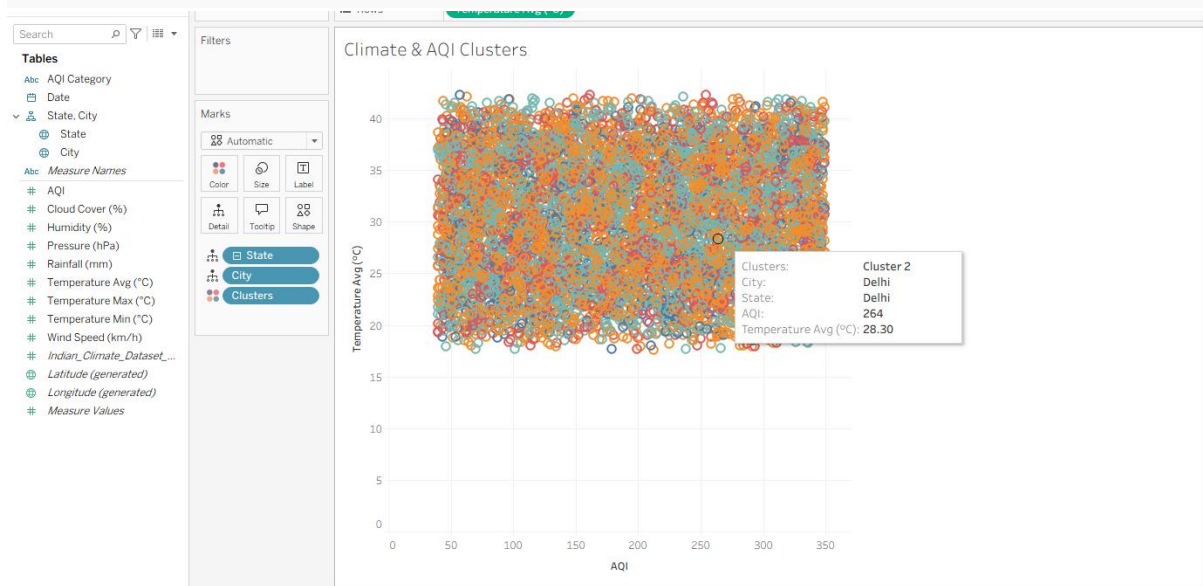
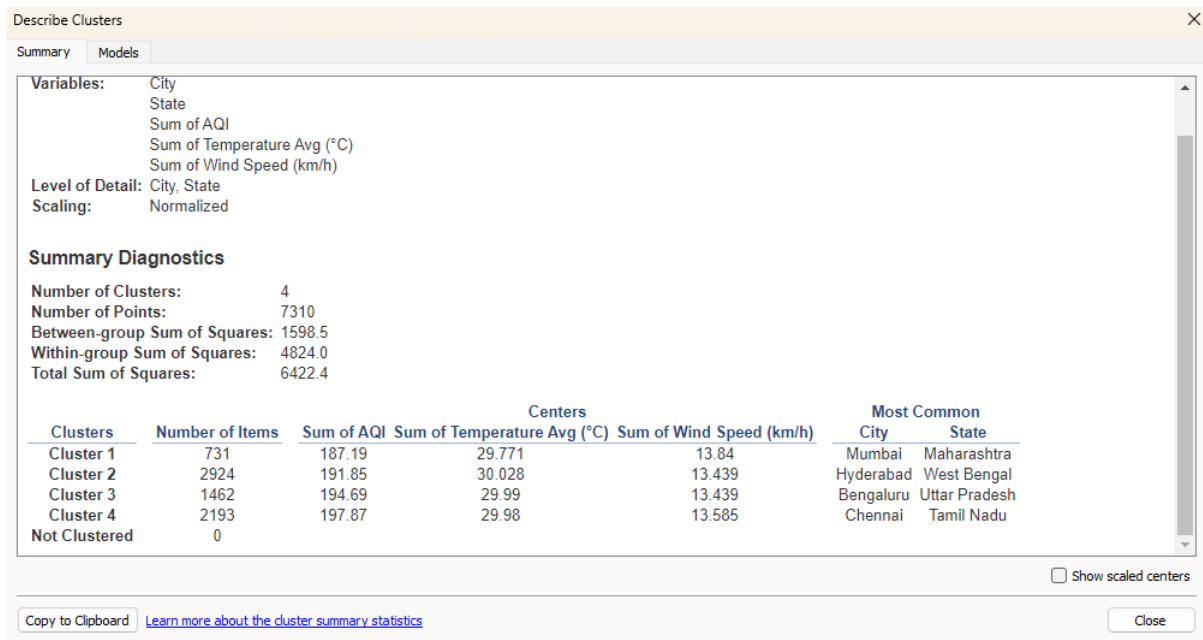




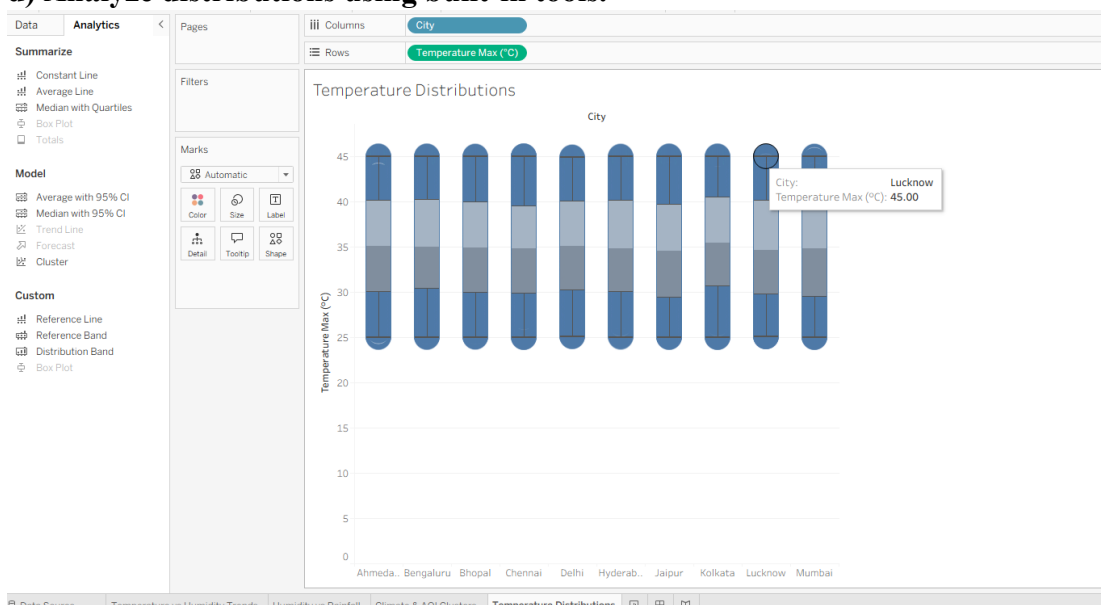
c) Perform clustering on datasets.

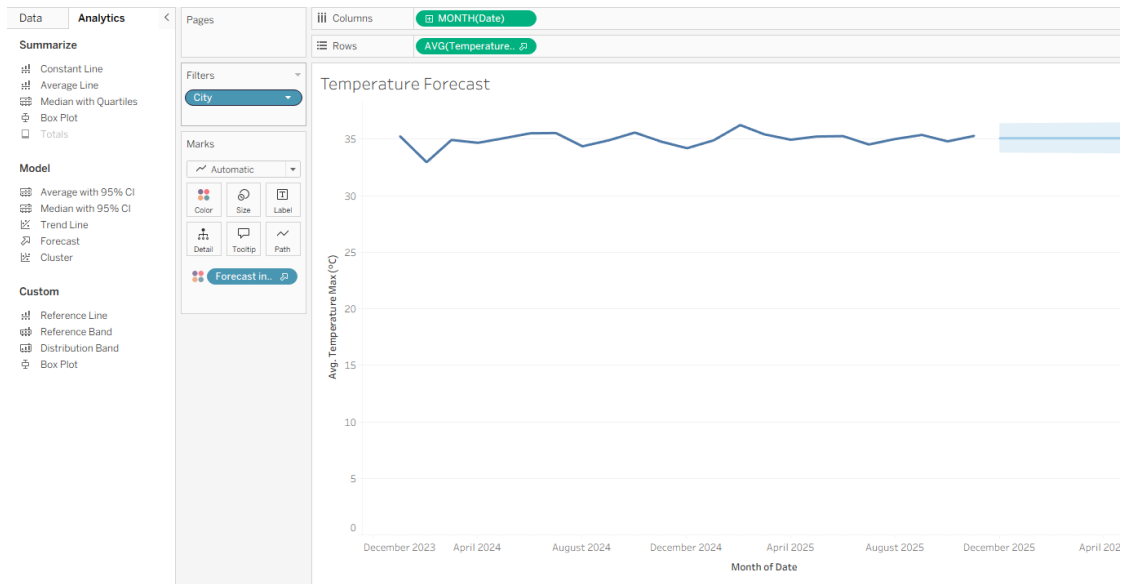
Describe Clusters						
Summary						
Models						
Analysis of Variance:						
Variable	F-statistic	p-value	Model		Error	
			Sum of Squares	DF	Sum of Squares	DF
Sum of AQI	3.377	0.01754	0.8442	3	608.8	7306
Sum of Wind Speed (km/h)	0.8725	0.4545	0.2132	3	595.1	7306
Sum of Temperature Avg (°C)	0.3618	0.7806	0.06336	3	426.5	7306

Categorical variables are not included in the Analysis of Variance table.

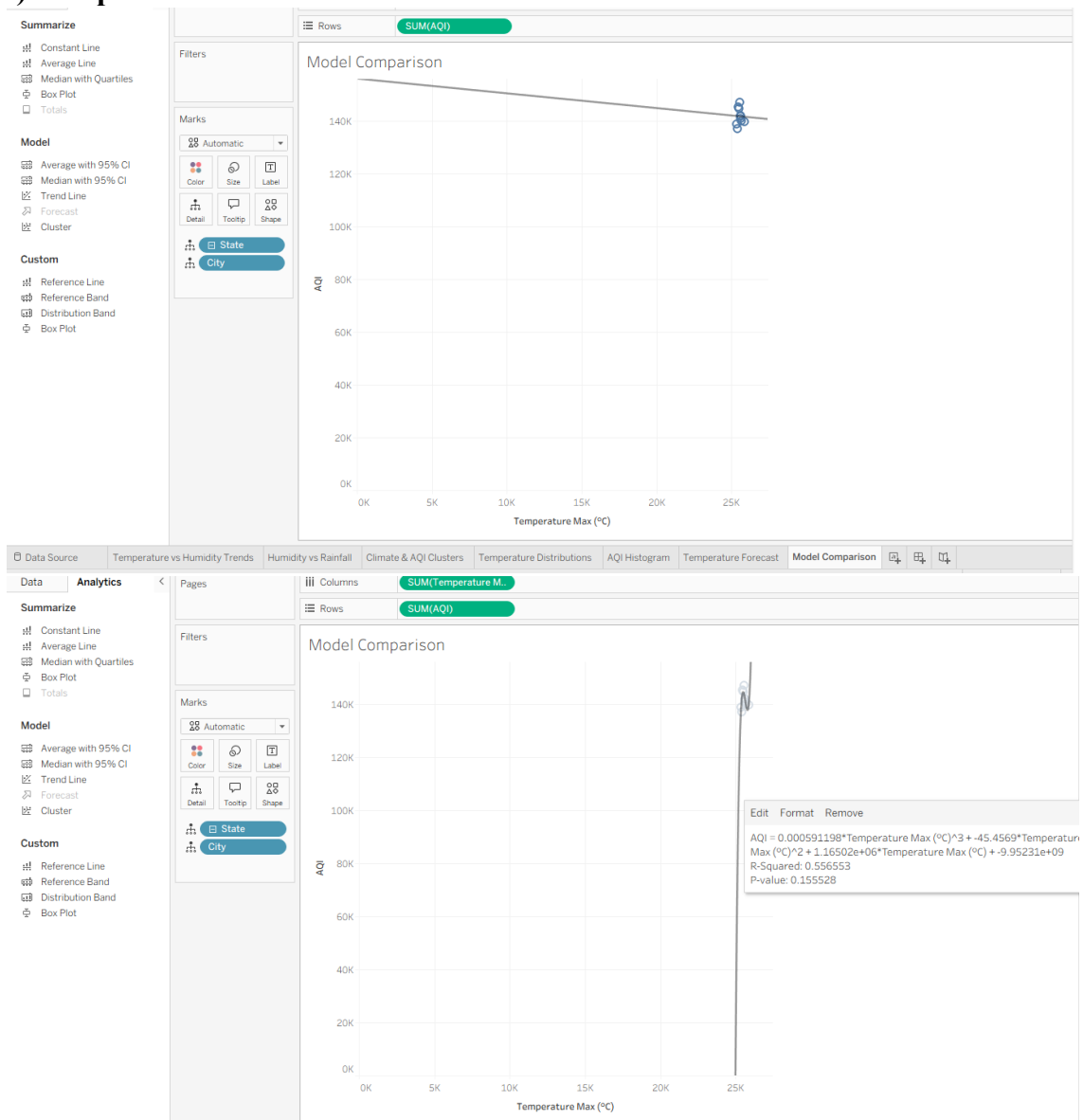


d) Analyze distributions using built-in tools.





f) Compare different statistical models.



g) Export and interpret statistical results

Describe Trend Line

P-value: 0.155528

Equation: $AQI = 0.000591198 \cdot \text{Temperature Max } (^{\circ}\text{C})^3 + -45.4569 \cdot \text{Temperature Max } (^{\circ}\text{C})^2 + 1.16502e+06 \cdot \text{Temperature Max } (^{\circ}\text{C}) + -9.95231e+09$

Coefficients

Term	Value	StdErr	t-value	p-value
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Describe Trend Model

freedom (DF):

SSE (sum squared error): 4.0043e+07

MSE (mean squared error): 6.67384e+06

R-Squared: 0.556553

Standard error: 2583.38

p-value (significance): 0.155528

Individual trend lines:

Panels		Line		Coefficients			
Row	Column	p-value	DF	Term	Value	StdErr	t-value
AQI	Temperature Max ($^{\circ}\text{C}$)	0.155528	6	Temperature Max ($^{\circ}\text{C}$)^3	0.0005912	The trend line n estimation requ a near singular matrix inversion model statistics unsupported.	
				Temperature Max ($^{\circ}\text{C}$)^2	-45.4569		
				Temperature Max ($^{\circ}\text{C}$)	1.16502e+06		
				intercept	-9.95231e+09		

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